

STAVELY AAFC, AB, Canada

WMO: 715550

Lat: 50.183N

Lon: 113.883W

Elev: 1364

StdP: 85.98

Time Zone: -7.00 (NAM)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.5	-23.2	-30.1	0.3	-25.3	-26.6	0.4	-21.5	16.0	3.2	13.9	2.5	3.2	340	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	27.5	14.9	25.6	14.3	23.7	13.8	17.1	23.5	16.0	22.8	14.9	21.8	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.7	12.3	20.3	13.4	11.3	19.0	12.1	10.4	17.9	53.4	23.5	49.8	22.7	46.5	21.7	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.0	10.2	DB	-28.8	31.0	3.5	2.1	-31.3	32.5	-33.3	33.7	-35.3	34.9	-37.8	36.5	(4)
(5)			WB	-29.2	19.3	3.4	1.6	-31.7	20.4	-33.7	21.4	-35.6	22.3	-38.1	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-4.1	-4.5	-0.9	3.7	9.1	12.5	16.6	16.1	11.7	5.7	-0.1	-4.3	(6)	
	DBStd	9.76	8.49	7.58	7.14	5.56	4.98	3.58	3.60	4.00	5.44	5.61	6.79	7.78	(7)	
	HDD10.0	2423	437	406	337	198	78	14	2	4	43	154	305	444	(8)	
	HDD18.3	4854	695	639	595	439	288	177	77	90	207	393	553	702	(9)	
	CDD10.0	668	1	0	1	9	49	91	207	194	94	20	2	1	(10)	
	CDD18.3	55	0	0	0	0	1	3	23	21	7	0	0	0	(11)	
	CDH23.3	473	0	0	0	1	12	21	191	188	59	1	0	0	(12)	
	CDH26.7	91	0	0	0	0	1	3	39	39	9	0	0	0	(13)	
(14) Wind	WSAvg	4.6	4.9	4.3	4.9	4.9	4.6	4.4	4.2	3.9	4.2	4.6	5.1	4.9	(14)	
(15) Precipitation	PrecAvg	467	15	17	23	39	67	103	48	50	45	24	20	17	(15)	
	PrecMax	809	28	37	41	70	154	370	118	114	140	77	43	35	(16)	
	PrecMin	234	6	4	6	16	4	25	0	1	4	5	5	4	(17)	
	PrecStd	124	6	7	9	16	39	78	33	27	32	18	11	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	11.7	14.8	20.8	25.0	26.3	30.1	30.0	27.9	21.3	16.0	12.2	(19)	
		MCWB	4.0	3.3	5.8	8.6	11.8	15.9	15.8	14.4	13.3	9.5	6.5	3.3	(20)	
	2%	DB	9.4	8.8	12.0	17.2	22.3	22.9	27.5	27.5	25.0	18.4	12.6	8.7	(21)	
		MCWB	2.8	2.4	4.4	7.4	11.0	13.8	15.8	14.4	12.4	8.9	5.5	2.3	(22)	
	5%	DB	7.2	6.5	9.9	14.5	19.6	21.0	25.5	25.5	22.5	15.6	10.3	6.6	(23)	
		MCWB	1.7	1.1	3.4	5.9	10.1	13.0	15.4	14.0	12.0	7.6	4.1	1.4	(24)	
	10%	DB	5.5	4.6	7.7	11.8	17.2	19.2	23.4	23.3	20.0	13.2	8.2	4.8	(25)	
		MCWB	0.8	-0.2	2.2	4.6	8.8	11.9	14.8	13.7	11.0	6.3	2.9	0.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	4.4	6.6	9.3	13.1	17.2	19.3	17.4	14.9	11.1	7.8	4.5	(27)	
		MCDB	11.0	10.1	13.4	19.0	22.2	23.5	25.5	24.5	23.3	18.8	13.8	10.3	(28)	
	2%	WB	3.2	2.7	5.0	7.7	11.7	15.1	17.7	16.3	13.6	9.5	6.0	3.0	(29)	
		MCDB	8.2	8.2	11.0	15.8	20.2	21.2	23.8	23.7	22.8	16.9	12.1	8.0	(30)	
	5%	WB	2.1	1.4	3.7	6.4	10.5	13.7	16.7	15.3	12.4	8.0	4.5	1.6	(31)	
		MCDB	6.6	6.2	9.3	13.6	18.3	19.7	22.9	22.7	21.1	14.9	9.7	6.2	(32)	
	10%	WB	1.0	0.1	2.4	5.0	9.3	12.5	15.6	14.3	11.3	6.7	3.0	0.3	(33)	
		MCDB	5.4	4.3	7.5	11.3	16.2	17.9	22.0	21.4	19.3	12.9	8.0	4.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.1	9.1	9.5	9.8	9.3	10.7	10.9	10.2	9.2	8.3	8.4	(35)	
		MCDBR	9.2	10.8	11.0	12.8	12.7	11.3	12.8	13.4	12.8	11.5	10.2	9.6	(36)	
	5% WB	MCWBR	5.8	6.7	6.3	6.7	6.4	6.5	6.5	6.0	6.0	6.0	6.1	6.2	(37)	
		MCWBR	8.7	10.9	10.9	12.2	11.4	10.4	11.0	11.6	11.9	11.0	10.2	9.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.241	0.253	0.272	0.319	0.325	0.323	0.324	0.338	0.301	0.269	0.247	0.225	(40)	
		taud	2.456	2.471	2.482	2.363	2.411	2.458	2.466	2.419	2.506	2.570	2.522	2.463	(41)	
	Ebn at Noon		838	914	950	927	932	934	927	896	903	884	827	813	(42)	
		Edh at Noon	57	73	87	110	110	106	104	104	85	66	53	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.18	2.16	3.37	4.72	5.58	6.07	6.77	5.48	3.86	2.38	1.28	0.91	(44)	
	RadStd		0.09	0.18	0.26	0.33	0.44	0.44	0.41	0.37	0.37	0.25	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page